The documentation and process conversion measures necessary to comply with this revision shall be completed by 30 January 1995

INCH-POUND

MIL-S-19500/427E
31 August 1994
SUPERSEDING
MIL-S-19500/427D
30 September 1993

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER,
TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622,
1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL,
1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US,
JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, power rectifier diodes. Four levels of product assurance are provided for each device as specified in MIL-S-19500. Two levels of product assurance are provided each unencapsulated device type.

1.2 Physical dimensions. See figures 1, 2, 3 (D-5A), 4, and 5 (JANHC and JANKC).

1.3 Maximum ratings.

Types	V_R	V_{RWM}	I_O T_A = +55°C 1/ 2/	I_O T_A = +100°C 2/ 3/	t_{rr}	T_{STG} and T_J	I_{FSM} T_A = +100°C I_O = 750 mA dc t_p = 8.3 ms	$R_{\theta JL}$ 4/
	V dc	V (pk)	A dc	mA dc	µs	°C	A (pk)	
1N5614, UL,US	200	200	1	750	2	-65 to +175	30	See figure 6
1N5616, 5/ UL,US	400	400	1	750	2	-65 to +175	30	
1N5618, 5/ UL,US	600	600	1	750	2	-65 to +175	30	
1N5620, 5/ UL,US	800	800	1	750	2	-65 to +175	30	
1N5622, 5/ UL,US	1,000	1,000	1	750	2	-65 to +175	30	

1/ From 1 A at T_A = +55°C, to 0.75 A dc at T_A = +100°C, derate linearly at 5.56 mA/°C.
2/ For the 1 A rating at the +55°C and the 750 mA rating at +100°C, no special mounting, heat sinking, or forced air flow across exposed areas of the device is required.
3/ From 0.75 A at T_A = +100°C, to 0 A dc at T_A = +175°C, derate linearly at 10 mA/°C.
4/ $R_{\theta JL} \leq 38°C/W$ at L = .375 inch (9.53 mm) for UL and US suffix type devices, $R_{\theta JEC}$ = 7.0°C/W.
5/ Barometric pressure reduced: 1N5616 and 1N5618 = 8 mm Hg, 1N5620 and 1N5622 = 33 mm Hg (not for UL and US suffix type devices).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Research Laboratory, ATTN: AMSRL-EP-RD, Fort Monmouth, NJ, 07703-5601 by using the Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification For.

STANDARDS

MILITARY

MIL-STD-750 - Test Method for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions shall be as specified in MIL-S-19500 and herein.

D.U.T. - - - - Device under test.

JANH - - - - High reliability product assurance level for unencapsulated devices.

JANK - - - - Space reliability product assurance level for unencapsulated devices.

T_{EC} - - - - Temperature of end-caps (mounting pads for US suffix devices).

US - - - - Unleaded or surface mounted diodes (square end-caps).

UL - - - - Unleaded or surface mounted diodes (square end-caps).

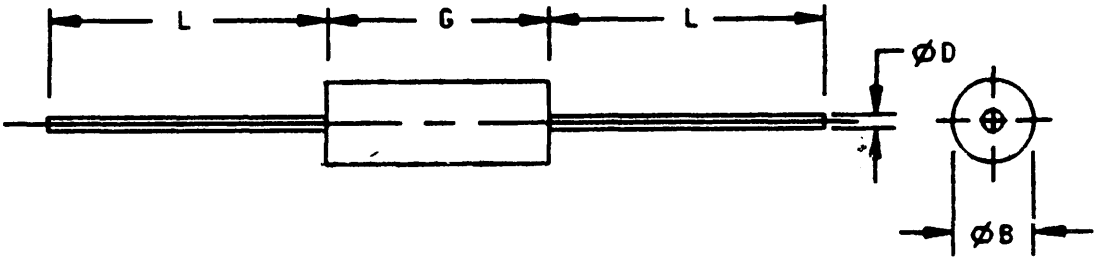
3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figures 1, 2, 3, 4, and 5 herein. Except for JANHC and JANKC, all devices shall be metallurgically bonded-thermally-matched-noncavity-double plug construction, utilizing a category I bond, in accordance with MIL-S-19500. UL and US version devices shall be structurally identical to the nonsurface mount version devices except for leaded configuration.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-S-19500, and as specified herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

3.4.1 Marking of UL and US version devices. For UL and US version devices only, all marking (except as stated in 3.5) may be omitted from the body, but shall be retained on the initial container.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

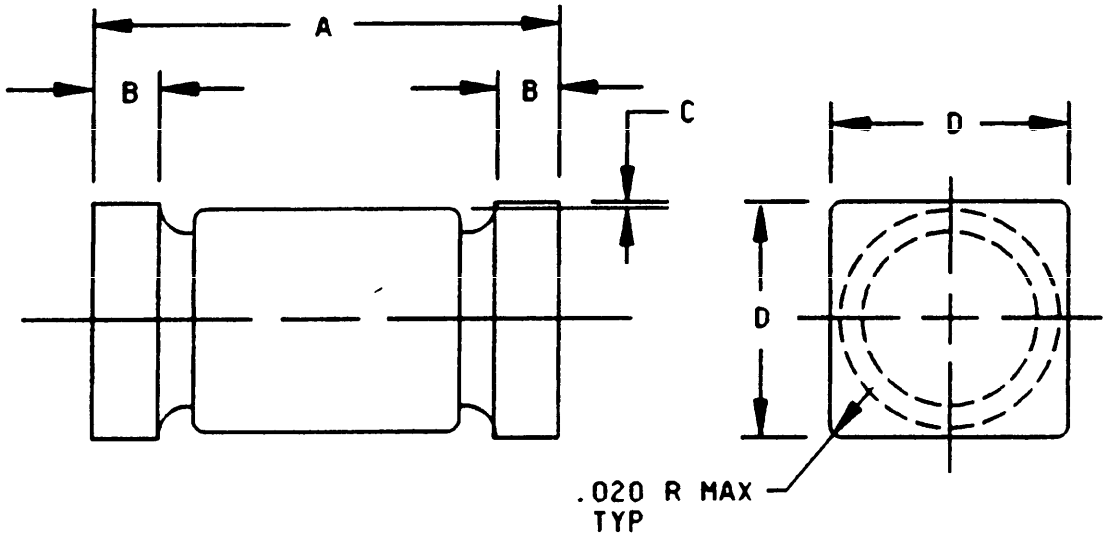


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
ØB	0.065	0.110	1.65	2.79	3
ØD	0.026	0.033	0.66	0.84	
G	0.130	0.225	3.30	5.72	4
L	1.00	1.30	25.4	33.02	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Dimension ØB shall be measured at the largest diameter.
4. The G dimension shall include all uncontrolled areas of the device.

FIGURE 1. Physical dimensions (for non-UL and non-US suffix devices only).

Source: https://assist.dla.mil -- Downloaded: 202610-18T16:55Z
Check the source to verify that this is the current version before use.

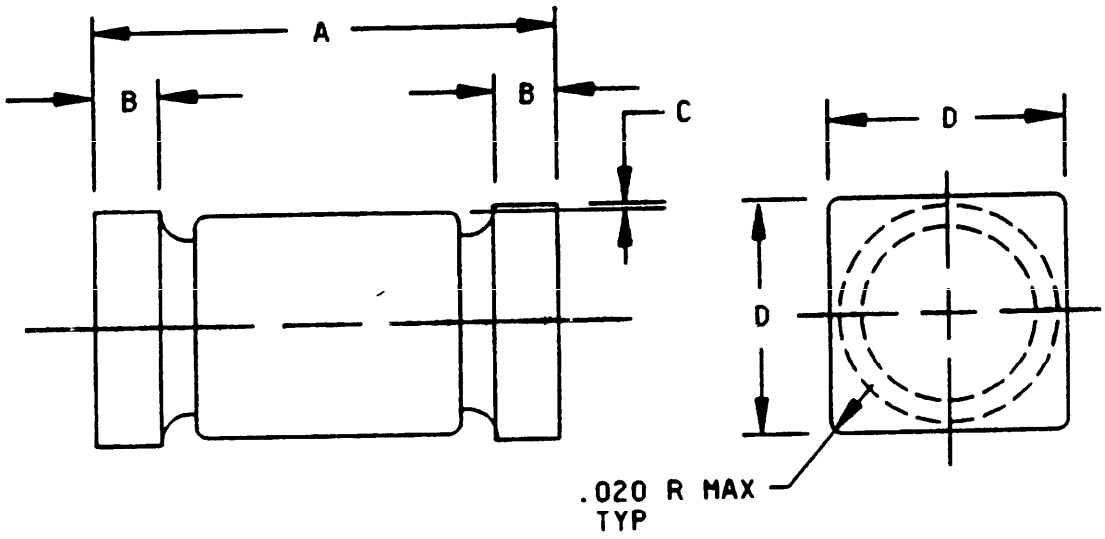


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.168	.225	4.27	5.72	
B	.019	.028	0.48	0.71	
C	.003	---	0.08	---	
D	.091	.125	2.31	3.18	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions types 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, and 1N5622UL (surface mount devices).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

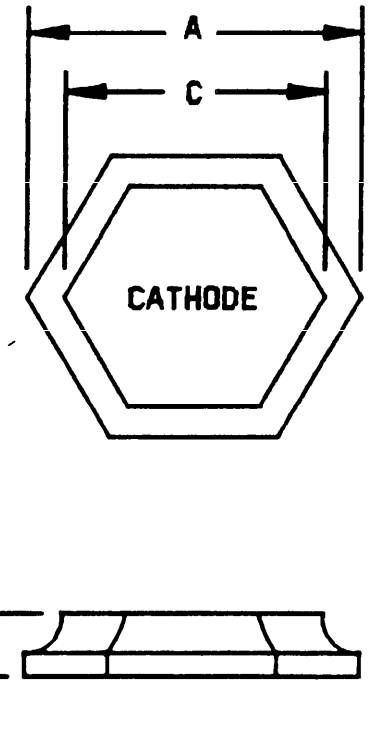


Symbol	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A	.168	.200	4.27	5.08	
B	.019	.028	0.48	0.71	
C	.003	----	0.08	----	
D	.091	.103	2.31	2.62	

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions types 1N5614US, 1N5616US, 1N5618US, 1N5620US, and 1N5622US (surface mount devices).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.



A - version

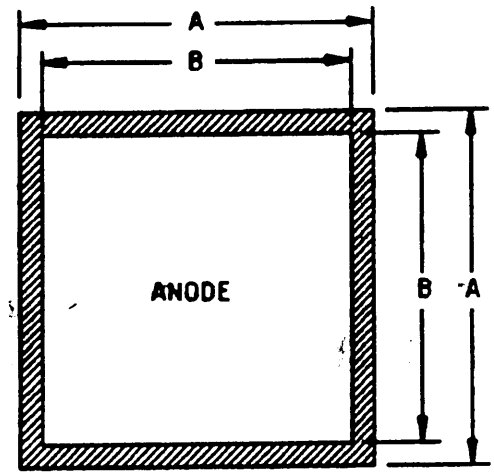
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.047	.053	1.19	1.35
B	.007	.011	0.18	0.28
C	.033	.037	0.84	0.94

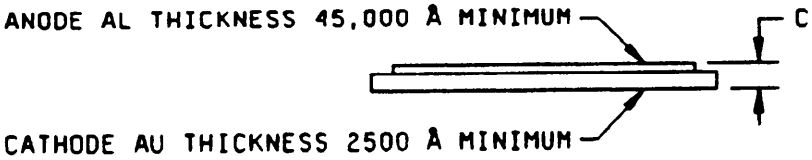
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal: Gold 10,000 Å minimum.
Back metal: Gold 4,000 Å minimum.

FIGURE 4. Physical dimension, JANHCA AND JANKCA die.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.





B - version

Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
A	.039	.043	1.00	1.09
B	.021	.025	0.53	0.64
C	.008	.012	0.20	0.31

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The physical characteristics of the die are:

Top metal:	Aluminum (anode)	45,000 Å minimum.
Back metal:	Gold (cathode)	2,500 Å minimum.

FIGURE 5. Physical dimension, JANHCB AND JANKCB die.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

3.5 Polarity. The polarity of all types shall be indicated with a contrasting color band to denote the cathode end. Alternatively, for UL and US suffix devices, a minimum of three contrasting color dots spaced around the periphery on the cathode end or a contrasting colorband may be used.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500.

4.2.1 JANHC and JANKC devices. Qualification for die shall be in accordance with appendix H of MIL-S-19500.

4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Thermal impedance (see 4.3.3)	Thermal impedance (see 4.3.3)
9	I_{R1} and V_F	Not applicable
11	I_{R1} and V_F, $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta_{VF} \leq$ ±0.1 V dc.	I_{R1} and V_F
12	See 4.3.1	See 4.3.1
13 2/	Subgroups 2 and 3 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq$ ±0.1 V dc. Scope display evaluation (see 4.5.3).	Subgroup 2 of table I herein: $\Delta I_{R1} \leq$ 100 percent of initial reading or ±100 nA dc, whichever is greater. $\Delta V_F \leq$ ±0.1 V dc. Scope display evaluation (see 4.5.3).

1/ Thermal impedance shall be performed any time after sealing provided temperature cycling is performed in accordance with MIL-S-19500, screen 3 prior to this thermal test.

2/ $Z_{\theta JX}$ need not be repeated at step 13 if performed earlier in the screening sequence.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: Method 1038, condition B of MIL-STD-750, T_A = room ambient as designated in 4.5 of MIL-STD-750, V_R = full rated V_{RWM} (see 1.3); f = 50-60 Hz, I_O = 1.0 A dc minimum, T_{EC} = +100°C to +125°C for surface mount devices (UL and US version) (see 4.5.7).

4.3.2 Screening (JANHC or JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H.

4.3.3 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. Maximum limit = 4.5°C/W.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

4.3.3.1 Thermal impedance ($Z_{\theta JX}$ measurements) for initial qualification or requalification. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101 (read and record value $Z_{\theta JX}$) and shall be supplied to the qualifying activity on a 500 piece sample from the qualification lot prior to qualification approval.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. The following test conditions shall be used for $Z_{\theta JX}$ in a group A, subgroup 2 inspection:

a. I_H measuring current - - - - - - - - - - - 10 mA.

b. I_H forward heating current - - - - - - - - 10 A.

c. t_H heating time - - - - - - - - - - - - - 10 ms.

d. t_{MD} measurement delay time - - - - - - - 100 µs maximum.

The maximum limit for $Z_{\theta JX}$ in group A, subgroup 2 is $Z_{\theta JX}$ (maximum) = 4.5°C/W.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JANTX and JANTXV) of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ shall be performed after JANS group B4 only. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Conditions
B3	4066	I_{FSM} = 30 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_{RWM} = rated V_{RWM} (see 1.3); T_A = +100°C.
B4	1037	I_O = 1 A dc minimum; T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3); t_{on} = t_{off} = 3 minutes minimum, 2,000 cycles.
B5	1027	I_O = 1 A dc minimum; V_R = rated V_{RWM} (see 1.3); T_A = +150°C minimum, adjust I_F and T_A as required to give an average T_J = +275°C +0°, -35°C (see 4.5.7).
B6	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.4 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan = 10, end-points, see 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

4.4.2.2 Group B inspection, table IVb (JANTX and JANTXV) of MIL-S-19500.

Subgroup	Method	Conditions
B2	4066	I_{FSM} = 30 A (pk); 10 surges of 8.3 ms each at 1 minute intervals, superimposed on I_O = 750 mA dc; V_R = rated V_{RWM} (see 1.3); T_A = +100°C.
B3	1027	T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1).
B5	3101 or 4081	+25°C ≤ T_A ≤ +35°C (recorded before test is performed); $R_{\theta JL}$ (maximum) ≤ 38°C/W; L = .375 inch (9.53 mm) (see figure 6). For surface mount devices (UL and US version), $R_{\theta JEC}$ ≤ 7°C/W.
B7	---	Peak reverse power, see 4.5.4 and figure 7 herein. P_{RM} ≥ 500 W. Test shall be performed on each sublot; sampling plan = 10, end-points, see 4.4.2.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the applicable inspections of table I, group A, subgroup 2 and as specified in table II herein. $Z_{\theta JX}$ need not be performed. See subgroup conditions for delta limits when applicable. Leaded samples may be used in lieu of surface mount devices (UL and US version) for life tests.

4.4.3.1 Group C inspection, table V of MIL-S-19500.

Subgroup	Method	Conditions
C2	2036	Tension: Test condition A, weight = 12 pounds. Lead fatigue: Test condition E, weight = 1 pound. NOTE: Both tension and lead fatigue are not applicable for surface mount devices (UL and US version).
C6	1026	T_A = room ambient as defined in the general requirement of MIL-STD-750 (see 4.5); I_O = 1 A dc minimum; f = 50-60 Hz; V_R = rated V_{RWM} (see 1.3, 4.5.7, and 4.5.7.1).

4.4.4 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-S-19500.

4.4.4.1 Group E inspection, table VII of MIL-S-19500. Subgroups E1 through E4 is not applicable.

Subgroup	Method	Conditions
E5	1001	1N5616, 1N5618 = 8 mm Hg (100,000 ft); 1N5620, 1N5622 = 33 mm Hg (70,000 ft); V_R = V_{RWM} (see 1.3); t = 1 minute (minimum).

4.5 Methods of inspection. Methods of inspection shall be specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Inspection conditions. Unless otherwise specified, all inspections shall be conducted at an ambient temperature (T_A) of +25°C ±3°C.

4.5.3 Scope display evaluation. The reverse breakdown characteristics shall be viewed on an oscilloscope with display calibration factors of 5 to 20 μA/division and 50 to 200 V/division. Reverse current over the knee shall be at least 50 μA. Each device may exhibit a sharp knee characteristic and any discontinuity or dynamic instability of the trace shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

hours (see 4.3.4)

4.5.4 Peak reverse power test. This test shall be measured in the circuit of figure 7, or equivalent. A 20 microsecond half-sine waveform of current shall be used and peak reverse power shall be determined by the product of peak reverse voltage and peak reverse current.

4.5.6 Thermal resistance. Thermal resistance measurement shall be performed in accordance with MIL-STD-750, method 3101 or 4081. Forced moving air or draft shall not be permitted across the device during test. The maximum limit for $R_{\theta JL}$ under these test conditions shall be $R_{\theta JL(max)}$ = 38°C/W at .375 inch (9.53 mm) and 7°C/W at 0 inch (0.00 mm).

4.5.7 Life tests. These tests shall be conducted with a half-sine waveform of the specified peak voltage impressed across the diode in the reverse direction followed by a half-sine waveform of the specified average rectified current. The forward conduction angle of the rectified current shall be neither greater than 180° nor less than 150°.

4.5.7.1 Mounting conditions. At the option of the manufacturer, any clips or heat sink mounting configurations may be utilized provided that I_o is adjusted such that the junction temperature of each diode is maintained at least +120°C above the specified ambient temperature of +25°C.

Source: https://assist.dla.mil -- Downloaded: 202610-18T16:55Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection.

Inspection	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Forward voltage	4011	I_F = 3 A dc, pulsed (see 4.5.1) t_p = 300 µs, 2 percent maximum duty cycle	V_F	0.8	1.3	V dc
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R1}		0.5	µA dc
Breakdown voltage	4021	I_R = 50 µA dc	$V_{(BR)R}$			
1N5614, UL,US				220		V dc
1N5616, UL,US				440		V dc
1N5618, UL,US				660		V dc
1N5620, UL,US				880		V dc
1N5622, UL,US				1100		V dc
Thermal impedance	3101	See 4.4.1	$Z_{\theta JX}$		4.5	°C/W
Subgroup 3						
High temperature operation		T_A = +100°C				
Reverse current leakage	4016	DC method; V_R = rated V_R (see 1.3)	I_{R2}		25	µA dc
Subgroup 4						
Reverse recovery time	4031	Condition B1.	t_{rr}		2	µs
Subgroups 5, 6, and 7						
Not applicable						

1/ For sampling plan, see MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

TABLE II. Groups A, B, and C electrical measurements. 1/ 2/ 3/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1	Reverse leakage current change	4016	DC method	ΔI_{R1} 4/		For JANTX and JANTXV, ≤ 250 nA dc or 100 percent, whichever is greater; for JANS, ≤ 100 nA dc or 100 percent, whichever is greater.	
2	Forward voltage change	4011	I_F = 3 A dc; Pulsed (see 4.5.1)	ΔV_{F1} 4/		±50 mV dc maximum change from previous measured value.	

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 2.

b. Subgroup 4, see table II herein, step 2.

c. Subgroup 5, see table II herein, steps 1 and 2.

2/ The electrical measurements for table IVb (JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, step 1.

b. Subgroup 6, see table II herein, step 1.

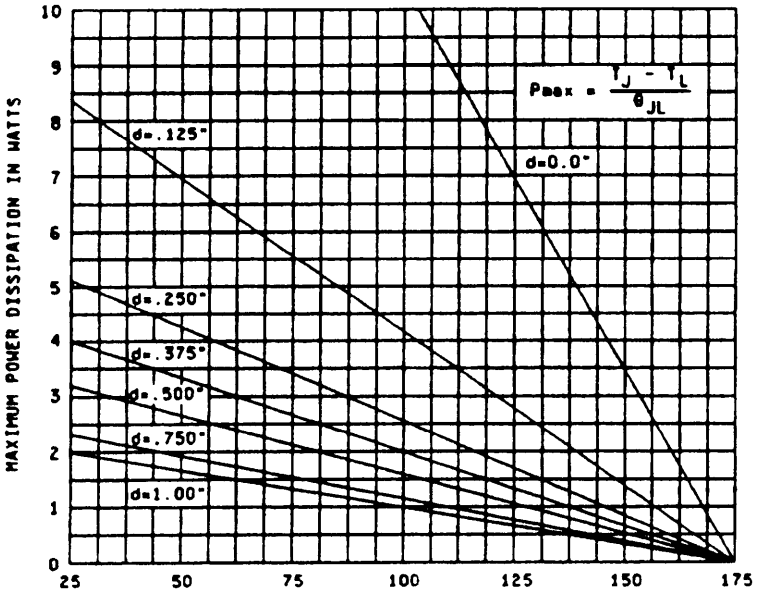
3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 2 (JANS).

b. Subgroup 6, see table II herein, steps 1 and 2 (JANS), step 1 (JANTX and JANTXV).

4/ Devices which exceed the group A limits for this test shall not be accepted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.



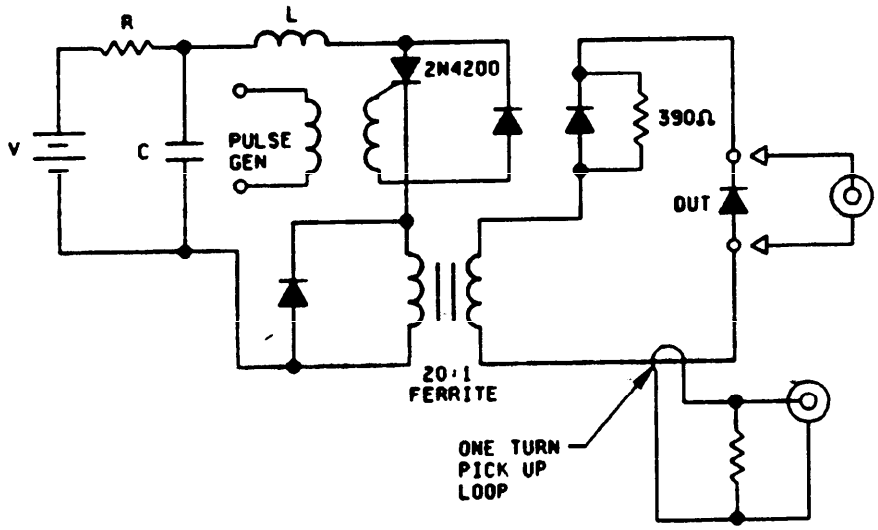
Maximum lead temperature in °C (T_L at point d from body (see table below)).
For maximum operating junction temperature of +175° with equal two-lead conditions.

d (see notes 1 and 2)	$R_{\theta JL}$
Dimensions	°C/W
0.000 (0.00 mm)	7
0.125 (3.18 mm)	18
0.250 (6.35 mm)	29
0.375 (9.53 mm)	38
0.50 (12.7 mm)	45
0.75 (19.1 mm)	62
1.00 (25.4 mm)	76

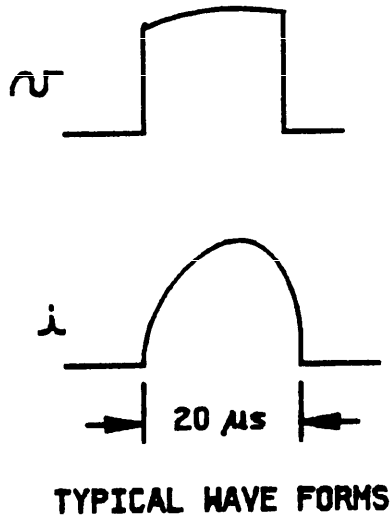
1. Lead distance point d and thermal resistance data.
2. Thermal resistance test performed at .375 inches (9.53 mm).
3. For US suffix devices: $R_{\theta JEC}$ = 7.0°C/W.

FIGURE 6. Power derating curves maximum power in watts versus lead temperature.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.



NOTES:
1. L - 13T #22 on 1 inch (25.4 mm) diameter form (air core).
2. C - 1 to 10 µfd to give 20 µs pulse width.
3. V - adjustable to 200 volts for power desired in D.U.T.



TYPICAL WAVE FORMS

FIGURE 7. Peak reverse power measurement circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory).

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Issue of DODISS to be cited in the solicitation.

b. Lead finish as applicable (see 3.3.1).

c. Product assurance level, type designator, and for die acquisition the JANHC and JANKC identification (see 6.4 and figures 4 and 5).

6.3 Supersession data. The following MIL-S-19500/427 types supersede MIL-S-19500/365 types:

MIL-S-19500/427 devices	MIL-S-19500/365 devices
1N5614	1N4383
1N5616	1N4384
1N5618	1N4385
1N5620	1N4585
1N5622	1N4586

6.4 Suppliers of die. The qualified die suppliers with the applicable letter version (e.g., JANHCA1N5614) will be identified on the Qualified Products List (QPL).

JANHC and JANKC ordering information		
PIN	Manufacturer	
	14552	33178
1N5614	JANHCA1N5614	JANHCB1N5614
	JANKCA1N5614	JANKCB1N5614
1N5616	JANHCA1N5616	JANHCB1N5614
	JANKCA1N5616	JANKCB1N5614
1N5618	JANHCA1N5618	JANHCB1N5614
	JANKCA1N5618	JANKCB1N5614
1N5620	JANHCA1N5620	JANHCB1N5614
	JANKCA1N5620	JANKCB1N5614
1N5622	JANHCA1N5622	JANHCB1N5614
	JANKCA1N5622	JANKCB1N5614

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5961-1609)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T16:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/427E	2. DOCUMENT DATE (YYMMDD) 940831

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFIER, TYPES 1N5614, 1N5616, 1N5618, 1N5620, 1N5622, 1N5614UL, 1N5616UL, 1N5618UL, 1N5620UL, 1N5622UL, 1N5614US, 1N5616US, 1N5618US, 1N5620US, 1N5622US, JAN, JANTX, JANTXV, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME DIRECTOR, US ARMY RESEARCH LAB/EPSD	b. TELEPHONE (Include Area Code) (1) Commercial: 908-544-3441/3148 (2) AUTOVON: 995-3441/3148
c. ADDRESS (Include Zip Code) ATTN: AMSRL-EP-RD FORT MONMOUTH, NJ 07703-5601	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 202610-18T16:55Z
Check the source to verify that this is the current version before use.